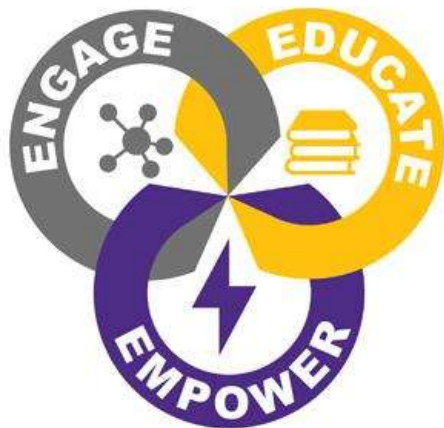


Tennessee Tech University
Curriculum & Instruction
HRED 7160-500, Fundamentals of Data Science
Online, 3 Credit Hours, Fall 2026



Mission

The mission of the College of Education and Human Sciences is to provide unique and rigorous, learner-centered academic experiences for our students to achieve their highest potential as life-long learners, professionals, and citizens.

Vision

The college will be an empowering force in education and human sciences.

Conceptual Framework

The College of Education and Human Sciences will prepare effective, engaging professionals through a clinically rich, evidence-based program with a network of mutually beneficial partnerships.

Instructor Contact Information

Feel free to contact me via email, Teams chat, office phone, or office visit.

Instructor's Name:

Michael Littrell, PhD

Office:

Bartoo 119

Office Phone

931-372-3207

Campus Email

mlittrell@tnitech.edu

Communication Details

Emails/messages

Please include your name and the course you are taking with me.

Course Information

Prerequisites

Completion of the courses (EDU-7300, EDU 7430) is a prerequisite for taking this course.

Required Texts

[A Hands-On Introduction to Data Science – Shah, 2020](#), first edition (ebook)

[Important!] - The textbook should be the First Edition for compatibility.

Recommended Texts

[Publication Manual of the American Psychological Association, Seventh Edition \(2020\)](#)

We will adhere to APA 7 guidelines in our writing.

Required References

All supplemental articles and material will be distributed from iLearn within the context of the course work.

Course Welcome and Description

Greetings All,

Welcome to Fundamentals of Data Science (HRED-7160), a gentle, hands-on introduction to data science in higher education that sets us up for the remaining three courses in the sequence. This course is a survey designed for everyone, whether you're new to data or an experienced professional. We'll briefly explore many topics across the data science workflow, including data wrangling, visualization, basic modeling, and interpretation. We'll put the "cart before the horse" a bit, arriving at meaningful results now and unpacking the details of the journey more systematically in the courses ahead. I find data science fascinating, and I hope by the end you'll want to keep learning as you apply the subject toward your professional goals.

Kindly,

Dr. Littrell

Course Objectives/Student Learning Outcomes

By the end of the course students will:

Outcomes table	
Outcome	Assignment/Assessment
Have a solid understanding of the fundamentals of data science as a subject.	The entirety of the course, execution of code and assignments, supporting notes from the text, weekly hands-on activities.
Have knowledge of how universities are using data science and the types of answers/outcomes that can be pursued for stakeholders.	Modules on higher education data science specifically, along with additional readings and reports on peer reviewed papers on the subject.
Know how to use various applications and languages to wrangle data, conduct exploratory data analysis, and apply data science workflows including machine learning techniques.	Modules on exploratory techniques, solving data problems, R, SQL, SPSS, supervised/unsupervised machine learning.
Understand how to work across the data science project life cycle including pre-planning, data ingest, collection, wrangling, modeling, deployment, iteration, communication, project management, and stakeholder management.	Modules and assignments about data science projects that define their parts, flow, and processes.

Major Teaching Methods

Asynchronous Course Content (iLearn)- This course will be conducted online asynchronously in Canvas where you complete weekly hands-on assignments, documenting the effort.

Synchronous Logistics Meetings (Teams) - We will have brief meetings (~30 minutes) via Teams to discuss the modules each week on Mondays at 5PM CT. Attendance is not required. They will be recorded. Here is the link to our meet every week:

<https://teams.microsoft.com/meet/28075007202206?p=Moa0HX4Pl0Mpc2H2C4>

Individual Meets – Meetings to discuss code or other content matters are scheduled as needed and most welcome.

Canvas Announcements – I will communicate important information via Canvas announcements.

Special Instructional Platform/Materials

Computer

You will need a [Mac or PC that meet the minimum specs per the college.](#)

Software

SPSS, R, and R studio are required. All applications used in the course are free or are no cost to students. See the install guide in the resources section of Canvas.

[Important!] Software should be installed immediately when the course starts to allow time to resolve any issues you may encounter.

Topics to be Covered

1. Introduction to data science
2. Data and its properties
3. Data science and its applications in higher education
 - a. Data science project management
4. Exploratory data analysis and statistical techniques
 - a. SPSS and stat fundamentals review
5. R and R studio for data science
6. Structured Query Language (SQL) in data science
 - a. Duck database creation
7. Introduction to machine learning
8. Supervised machine learning methods
 - a. Logistic Regression
 - b. SoftMax Regression
 - c. K Nearest Neighbor (KNN)
 - d. Decision Trees
 - e. Random Forest
 - f. Naive Bayes
 - g. Support Vector Machines
9. Unsupervised machine learning methods
 - a. Agglomerative Clustering
 - b. Divisive Clustering (K-Means)
 - c. Expectation Maximization
 - d. Reinforcement Learning
10. Solving data problems
11. Data collection methods

Course Schedule

[Important!] The applied code guide (ACG) is due by midnight on the last day of the course.

Schedule table							
W	Opn	Cls	N	Off	Detail	Mod	Topic
1	8/20	8/30	11			1	Intro
2	8/31	9/6	7			2	Data Preprocess
3	9/8	9/13	6	9/7	Lab. D	3	Higher Ed Data
4	9/14	9/20	7			4	Techniques
5	9/21	9/27	7			5	R
6	9/28	10/4	7			6	SQL
7	10/5	10/11	7			7	Intro to ML
8	10/12	10/18	5	10/15-16	Fall B	8	Supervised ML
9	10/19	10/25	7			8	Supervised ML
10	10/26	11/1	7			9	Unsupervised ML
11	11/2	11/8	6	11/3	Civ D	9	Unsupervised ML
12	11/9	11/15	7			10	Solv. Data Prob.
13	11/16	11/22	7			11	Ind. Code Exploration
14	11/23	11/29	4	11/25-27			Collate Applied Code Guide
15	11/30	12/4	5				Submit Applied Code Guide

Course Breakdown

1. Each week, engage with the assigned module in iLearn.
 - a. The modules contain video lectures, informative web documents, hands-on exercises, and assignments.
 - b. Many modules correspond to a chapter in Shah to read.
2. Execute all the coding examples seen in the videos.
 - a. Executing code from the textbook is optional, **but generally not recommended.**
3. Document your coding journey per the instructions in the Applied Code Guide (ACG).
4. Complete small assignments and exercises as seen in the modules.
5. Collate all materials into the ACG and submit it on the last day of the course.

Grading and Evaluation Procedures

The Applied Code Guide will be 100% of your grade.

Grading Scale

Letter Grade	Grade Range
A	90-100
B	80-89
C	70-79
D	60-69
F	59 and below

Course Policies

Student Academic Integrity Policy

Maintaining high standards of academic integrity in every class is critical to the reputation of Tennessee Tech, its students, faculty, alumni, and the employers of Tennessee Tech graduates. Academic integrity is at the foundation of the educational process and key to student success. Students with academic integrity are committed to honesty, ethical behavior, and avoiding academic integrity violations. All students must read and understand Policy 216: Student Academic Integrity. Please see the Academic Integrity website (<https://www.tntech.edu/provost/academicintegrity/>) for more information.

Attendance Policy

This course is asynchronous with **no mandatory synchronous meetings** defined presently. Engaging with the content and submission of course assignments by the close dates are required.

Students who are unable to attend class for an extended period of time due to an emergency/extenuating circumstance (i.e., medical illness, hospitalization, death in the family/bereavement, military or legal obligation), may contact the Office of the Vice President for Student Affairs at studentaffairs@tntech.edu to request an absence notification.

Class Participation

Thorough and timely engagement with the course content and the resulting project submission are essential and will comprise all of your grade for the course.

Assignments and Related Policy

Late submissions should be requested in advance. Approval of extensions and whether full or partial credit will be given is at the professor's discretion.

I understand that emergencies do happen. If your submission extension is last minute, exceptions for submission of late work might be granted on a case-by-case basis. Supporting documentation (e.g., doctor's notes for illness) may be required.

The weekly hands-on nature of the course is meant to develop real-world skills that are built incrementally throughout the semester. For this reason, if a module is not completed during the time allotted, then you will still need to complete the work for your own learning progress.

More than 2 requests for extensions will require a discussion with me.

Disability Accommodation

Students with a disability requiring accommodations should contact the accessible education center (AEC). An accommodation request (AR) should be completed as soon as possible, preferably by the end of the first week of the course. The AEC is located in the Roaden University Center, room 112; phone 931-372-6119. For details, view Tennessee Tech's policy 340 – [services for students with disabilities at policy central](#).

Additional Resources

Technical Help

If you are experiencing technical problems, visit the [myTech IT Helpdesk](#) for assistance.

If you are having trouble with one of the instructional technologies (i.e. Zoom, Teams, Qualtrics, Respondus, or any technology listed [here](#)) visit the [Center for Innovation in Teaching and Learning](#) (CITL) website or call 931-372-3675 for assistance.

Tutoring

The university provides free tutoring to all Tennessee Tech students. Tutoring is available for any class or subject, as well as writing, test prep, study skills, and resume support. Appointments are scheduled, so contact the [Learning Center website](#) for more information.

Counseling and Health Services

Tennessee Tech offers support for student well-being through two key services. The Center for Counseling and Mental Health Wellness provides brief, solution-focused therapy to help students navigate personal and social challenges. Health Services delivers accessible, high-quality, and affordable medical care to promote overall wellness. Visit their respective websites to learn more or schedule an appointment.

Emergency Preparedness Protocols

Each student must take personal responsibility for following any University protocol related to pandemics, natural disasters, and other public health and safety events. Students are expected to follow all directives published by Tennessee Tech on its [Environmental Health & Safety webpage](#).

Generative AI: Open Use Guidelines

AI policy statement: Permitted in this Course with Attribution.

In this course, Generative AI resources are encouraged to be used by students with the purpose of supporting their work. To ensure academic integrity, students must openly disclose any AI-generated material they utilize and provide proper attribution. This includes in-text citations, quotations, and references.

To indicate the use of a Generative AI resources, a student should include the following statement in their assignments: "The author(s) acknowledge the utilization of [Generative AI Tool Name], a language model developed by [Generative AI Tool Provider], in the preparation of this assignment. The [Generative AI Tool Name] was employed in the following manner(s) within this assignment [e.g., brainstorming, grammatical correction, citation, specific section of the assignment]."

Proper citation guidelines can be found on the [CITL website](#).

Permitted Uses

Typically, GAI can be used in a thoughtful way to assist you in carefully thinking through your assignments incrementally and leveraged to scaffold your learning.

Examples

- Coding help where you ask about syntax, errors, structure, etc.
- Understanding code and how it functions.
- Requesting help with formatting and organization of code.

Prohibited Uses

Typically, GAI should not be used as a substitute for your own learning. As a rule of thumb, you should be able to clearly explain your code, methods, or results in your own words. Your code should relate to what you have learned in the course. Also, using AI to generate your narrative in the ACG is prohibited

Examples

- Copy-pasting AI-generated reports or code as-is into submissions.
- Copy-pasting AI-generated commented code sections.
- Copy-pasting AI analysis results interpretation.
- Using code proposed by AI that has no relevance to the content covered even if it functions.
- Using narrative prose in the ACG explaining work.